Runbook: running and managing this repository

Generated 2026-11-07 from docs/RUNBOOK.md · 295 lines of source

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How to set it up, run it, change it safely and keep it healthy. For how the pieces fit together read ARCHITECTURE.md; for deploying read DEPLOYMENT.md.

Cheat sheet

```
corepack enable && pnpm install        # once, and after pulling dependency changes
pnpm db:local                          # embedded Postgres with the shared schema (own terminal)

pnpm --filter @wadzzo/ar     dev -p 3000     # AR collector
pnpm --filter @wadzzo/brand  dev -p 3001     # brand portal
pnpm --filter @wadzzo/market dev -p 3002     # fan marketplace
pnpm --filter @wadzzo/express dev            # worker on :4000 (health only)

pnpm typecheck && pnpm lint && pnpm test && pnpm format:check    # the CI gate
pnpm env:check                                                    # env files agree
```

1. Prerequisites

Tool	Version	Notes
Node.js	`>=20.19` (`.nvmrc` says 22, the recommended one)	`nvm use` picks it up
pnpm	`12.9.1` (the `packageManager` field)	`corepack enable` installs the right one; do not use npm or yarn here
Git	any recent	
Postgres	optional	`pnpm db:local` gives you one with no install
Expo tooling	only for `apps/mobile`	it installs with **npm** on its own, see 8

2. First-time setup

```
git clone <repo> wadzzo-monorepo && cd wadzzo-monorepo
corepack enable
pnpm install                  # also runs `prisma generate` through @wadzzo/db
```

Then give each app you want to run its environment:

```
cp apps/brand/.env.example apps/brand/.env.local      # fill in; the file explains every variable
cp apps/market/.env.example apps/market/.env.local
cp apps/ar/.env.example    apps/ar/.env.local
cp apps/express-wadzzo/.env.example apps/express-wadzzo/.env
```

Real values are git-ignored (`.env` , `.env.*`); only `.env.example` is tracked. A teammate's real `.env.development` / `.env.production` files are shared out of band, never through git. See 5.

3. Running things locally

3.1 Database

Option	How	When
Embedded (`pnpm db:local`)	Postgres on `127.0.0.1:54329` , the schema from `packages/db` , and the two platform rows (`wadzzo` , `clintoncounty`). **In memory**: gone when you stop it.	quick checks, tests by hand, no network
A development database	`DATABASE_URL` in the app's `.env.local`	working with realistic data

For the embedded one use:

```
DATABASE_URL=postgresql://postgres:postgres@127.0.0.1:54329/postgres
DATABASE_POOL_MAX=1
```

It accepts **one connection at a time**, so run **one app at a time** against it.

3.2 The apps

Use fixed ports so the URLs in `.env.example` line up:

App	Command	Port	URL variables to match
AR (`web`)	`pnpm --filter @wadzzo/ar dev -p 3000`	3000	`NEXT_PUBLIC_URL` , `NEXTAUTH_URL` = `http://localhost:3000`
Brand (`brand`)	`pnpm --filter @wadzzo/brand dev -p 3001`	3001	`NEXT_PUBLIC_URL` , `NEXTAUTH_URL` = `http://localhost:3001` ; `NEXT_PUBLIC_WADZZO_AR_URL=http://localhost:3000`
Market (`app`)	`pnpm --filter @wadzzo/market dev -p 3002`	3002	`NEXT_PUBLIC_URL` , `NEXTAUTH_URL` = `http://localhost:3002` ; `NEXT_PUBLIC_BRAND_PORTAL_URL=http://localhost:3001`
Worker	`pnpm --filter @wadzzo/express dev`	4000	`DATABASE_URL` : the same database as the brand (its jobs table is how the two find each other)

Notes:

- `pnpm dev` (everything at once) works, but the apps then take whatever ports are free. Prefer the explicit commands above.
- Each app's `predev` hook runs `sync:platform`, which writes `src/styles/platform-theme.css` and `public/platform/**` for the current `PLATFORM_SLUG`. They are generated and git-ignored.
- AR runs on webpack (`--webpack`, already in its script); brand and market use Turbopack.
- The map agent, bulk pins and hotspot drops run in the worker. The brand app only adds jobs to the `jobs` table, so start the worker against the **same database** as the brand, or those jobs just wait. (Apply `packages/db/prisma/sql/2026-11-07-job-queue.sql` to a database that doesn't have the table yet.)

3.3 Another platform (Clinton County)

```
cd apps/brand
cp .env.clintoncounty .env.development.local     # overrides only; the rest comes from
.env.development
pnpm dev -p 3101
rm .env.development.local                         # back to Wadzzo
```

The theme, branding and assets follow `PLATFORM_SLUG`. Use a separate browser profile per platform: both share the `localhost` login cookie.

3.4 The mobile app

`apps/mobile` is **not** a workspace member and installs with npm (Metro wants a flat `node_modules`):

```
cd apps/mobile
npm install
npx expo start
```

It talks to the AR app's `/api/mobile/v1`. Copy `.env.example` to `.env` and set `EXPO_PUBLIC_API_URL` to your AR app (a phone needs your machine's LAN address, not `localhost`).

4. Everyday commands

Command	What it does
`pnpm typecheck`	`tsc` for every app and package (apps run `next typegen` first)
`pnpm lint`	ESLint; errors fail, warnings are a to-do list
`pnpm test`	all unit tests (Vitest, no network, no database)
`pnpm format` / `pnpm format:check`	Prettier for the whole repo
`pnpm db:generate`	regenerate the Prisma client after a schema edit
`pnpm db:studio`	Prisma Studio against `DATABASE_URL`
`pnpm env:check`	verifies the real env files (see 5)
`pnpm build`	every app and package (see 9 before running it locally)

Scope a command to one package (much faster):

```
pnpm --filter @wadzzo/api typecheck
pnpm --filter @wadzzo/api exec vitest run src/brand           # a folder
pnpm --filter @wadzzo/wallet exec vitest watch                 # watch mode
pnpm --filter @wadzzo/brand... typecheck                       # the app + its dependencies
```

Turbo caches `typecheck`, `lint` and `test`. After editing the Prisma schema run `pnpm db:generate` first or types will be stale.

5. Environment files

File (in `apps/<app>/`)	In git	Holds
`.env.example`	yes	every variable the app reads, grouped, with a description
`.env.local`	no	your machine's values (loaded in dev and prod mode)
`.env`, `.env.development`, `.env.production`	no	the shared real values (same sections as the example)
`.env.clintoncounty`, `.env.clintoncounty.production`	no	Clinton County overrides only (brand and AR)

Next.js loads, highest first: `.env.<mode>.local`, `.env.local`, `.env.<mode>`, `.env`. A variable already in the shell wins over all of them.

Check before you deploy or after any edit:

```
pnpm env:check            # both modes
pnpm env:check production
```

It never prints secrets (it compares hashes). It fails when something that must be identical across the apps differs (database, issuer key, platform wallets, bucket, Firebase), or when a production URL doesn't match the app's domain.

Adding an environment variable

1. Read it through the right door: API code via `packages/api/src/env.ts` (add the name to `REQUIRED` or `OPTIONAL`; it is read on first use, never at import); app code via the app's `src/env.js`.
2. Add it to every affected `apps/<app>/.env.example` with a one-line description.
3. If the apps must agree on it, add it to `MUST_MATCH` in `scripts/check-env.mjs`.
4. Set it in the Vercel projects / on the worker (DEPLOYMENT.md).

6. Database

- The schema lives **only** in `packages/db/prisma/schema.prisma`; every app and the worker generate their client from it.
- Import models and enums from `@wadzzo/db` (browser-safe) and the client `db` from `@wadzzo/db/server`.
- **Never** run `prisma db push` by habit. It can drop data on a shared database, so the script refuses unless you say so:

  ```
  CONFIRM_DB_PUSH=yes DATABASE_URL="<the database you mean>" pnpm --filter @wadzzo/db push
  ```

  Print `DATABASE_URL`'s host first. Production is changed from a trusted machine, additive changes first.

- Hand-written one-off SQL (new tables, backfills) lives in `packages/db/prisma/sql/`, named by date, applied in date order.
- Demo data: `packages/db/prisma/seed-clinton-demo.ts` and `seed-events.ts` (read their headers; they take `--clean`). AR has `pnpm --filter @wadzzo/ar seed:murals`.
- New rows of platform-scoped models **must set** `platformId`. A test (`packages/api/src/brand/__tests__/platform-stamping.test.ts`) scans every app's creates and fails if one is missing.

7. Making changes

Where code goes

You are adding	Put it in
A tRPC procedure or REST handler	`packages/api/src/<brand|market|ar>/routers/` (`rest/`, `mobile/`, `game/` for REST); apps only re-export
Logic two apps need	`packages/api/src/shared/` or the package that owns the concept
A form or request schema used by UI and API	`@wadzzo/schemas`
Pure AR types, distance, rarity, mural constants	`@wadzzo/ar-core` (no server imports, usable by the screens and mobile)
A UI component or page pattern	`@wadzzo/ui` (`components/` shadcn, `patterns/` shared layouts)
Wallet / signing / network config	`@wadzzo/wallet`
Branding, theme, assets for a platform	`@wadzzo/platform` (PLATFORMS.md)
A screen or page	the app (`apps/<app>/src/app` or `components`)

Rules (also in ARCHITECTURE.md): packages never import from an app or use `~/`; no `import "server-only"` in packages; settings come from `@wadzzo/api/env`; failures are `AppError` with an `ErrorCodes` value, never a bare `Error`; no duplicated helper between apps.

Recipes

A procedure

1. Create or extend a router under `packages/api/src/<app>/routers/`; build it with `createTRPCRouter`, `publicProcedure`, `protectedProcedure` from `@wadzzo/api` (brand also has `adminProcedure`, `creatorProcedure` in `brand/trpc.ts`).
2. Validate input with zod; throw `new AppError(ErrorCodes.X, { message })`.
3. Mount it in that app's root router (`<app>/router.ts` or the domain's `root.ts`).
4. A procedure that takes `signWith` is checked against the session automatically; do not add your own check.
5. Add a test next to it (`__tests__/`) using `createCallerFactory` with a stub `db`.

An error code: add it to `packages/errors/src/error-codes.ts` with its HTTP status, tRPC code and a safe default message.

A REST route: put the handler in `packages/api/src/<app>/{rest,mobile,game}/...`; the app's route file is one re-export (`export { GET } from "@wadzzo/api/ar/rest/..."`). Route-segment config (`dynamic`, `config.api.bodyParser`) must stay literal in the app's route file.

A schema change: edit `schema.prisma` → `pnpm db:generate` → typecheck → write a dated SQL file for production if data must move → apply as in 6.

A dependency: add the version to the `catalog:` in `pnpm-workspace.yaml`, then `"dep": "catalog:"` in the package that uses it. Do not put a version number in a package's own `package.json`. Packages that use it declare it themselves (nothing is hoisted for you).

A new package: copy `packages/geo` (`package.json`, `tsconfig.json`, `eslint.config.mjs`, `vitest.config.mts`), set `name`, `exports` and scripts, add `"@wadzzo/<name>": "workspace:*"` to the consumers, `pnpm install`.

A shadcn component: `cd packages/ui && pnpm dlx shadcn@latest add <name>`, then change the generated imports to the package's relative ones (`../lib/cn`) and run `pnpm format`.

8. Tests and quality gates

Layer	Where	Runs in CI	Needs
Unit	every package's `src/__tests__`, apps' `tests/`	yes	nothing (`SKIP_ENV_VALIDATION=1` is set)
End-to-end	`apps/{brand,market,ar}/e2e`, `pnpm test:e2e`	no	a dev database and Stellar testnet

The CI gate (`.github/workflows/ci.yml`) is: format check, typecheck, lint, unit tests. Run all four locally before a PR. Lint warnings are allowed; **new** warnings in code you touched are not welcome.

Commit and PR rules:

- Never commit `.env*` files other than `.env.example`.
- Do not add a `Co-Authored-By: Claude` trailer or any other AI attribution.
- Keep schema changes, dependency bumps and refactors in their own commits.

9. Building

```
pnpm --filter @wadzzo/brand build
```

Read this first:

- A build needs the `NEXT_PUBLIC_*` values and (unless you pass `SKIP_ENV_VALIDATION=1`) the whole validated environment. For a smoke build with no real env: `SKIP_ENV_VALIDATION=1 NEXT_PUBLIC_STELLAR_NETWORK=testnet pnpm --filter @wadzzo/ar build`. Market additionally needs `NEXT_PUBLIC_URL` and `NEXT_PUBLIC_BRAND_PORTAL_URL`.
- If `apps/<app>/.env.production` exists on your machine it **is loaded** by `next build`. Pin `DATABASE_URL` to something harmless (`postgresql://postgres:postgres@127.0.0.1:1/none`) so a build can never touch a real database.
- The market build prerenders `/sitemap.xml`, which queries the database. Without a reachable database it stops there after 18 of 19 pages; that is expected.
- `next build` and `next dev` in the same app share `.next`. Stop the dev server of that app first, or build in a separate checkout.

The worker builds with `pnpm --filter @wadzzo/express build` (tsup) and starts with `node dist/index.js`.

10. Keeping dependencies healthy

```
pnpm outdated -r            # what is behind
```

Bump the version in the `catalog:` of `pnpm-workspace.yaml`, `pnpm install`, then run the CI gate and the e2e tests of the apps that use it. pnpm may add `minimumReleaseAgeExclude` entries for very new releases; that is intended.

Held back on purpose:

Package	Why
`prisma` / `@prisma/client` 7.10	8 is a release candidate
`typescript` 6.0.x	`typescript-eslint` does not support 7 yet
the web3 stack (`@web3modal/*` , `@walletconnect/*` , Albedo, xBull, HOT)	pinned to what the wallets were verified with

TypeScript 6 does not include `@types/*` implicitly: a package that uses Node globals sets `"types": ["node"]` and lists `@types/node` .

11. Releasing and operating

- Branches: `main` (production), `develop` (development environment). Web apps deploy from Vercel on push; the worker from `.github/workflows/deploy-express-docker.yml` . Details, rollbacks and the checks after a deploy are in DEPLOYMENT.md.
- Before a release: CI green, `pnpm env:check production` , and for schema changes the SQL applied first.
- Secrets rotation: a value used by several apps (`ISSUER_ENCRYPTION_KEY` , `MOTHER_SECRET` , `NEXTAUTH_SECRET` , the database password) changes in **every** project and on the worker in one window, then `pnpm env:check` .

12. Troubleshooting

Symptom	Cause and fix
`Cannot find module '.prisma/client'` or stale model types	`pnpm db:generate`
`CONFIG_MISSING` naming a variable	That setting isn't set for this app; compare with its `.env.example` (API settings are read on first use, so it appears when a feature is used)
`Invalid environment variables` at start	The app's `src/env.js` rejected a value; the message names the variable
`This module cannot be imported from a Client Component module`	an `import "server-only"` reached a pages-router route; remove it from package code
The map agent or a pin batch never answers	the worker isn't running, or it uses a different `DATABASE_URL` than the brand
Port already in use	something is listening there; find it with `lsof -nP -iTCP:3001 -sTCP:LISTEN` and check before killing: it may be a teammate's or your own other terminal
Prisma connection errors with `pnpm db:local`	it accepts one connection: one app at a time, `DATABASE_POOL_MAX=1`
`tsc` says `Cannot find name 'process'` in a package	add `@types/node` and `"types": ["node"]` to that package (TypeScript 6)

Symptom	Cause and fix
Theme or logo looks like the wrong platform	`PLATFORM_SLUG` differs from the one the app last synced; restart `dev` so `sync:platform` reruns
Login redirects in a loop locally	`NEXTAUTH_URL` doesn't match the port you're on, or two platforms share the `localhost` cookie
A package change doesn't show up in an app	restart `next dev`; if the package is new and TS-only add it to `transpilePackages` in the app's `next.config.js`
`pnpm install` refuses a lockfile	someone edited a `package.json` without installing; run `pnpm install` and commit `pnpm-lock.yaml`
In zsh, commands vanish after a loop	don't use `path` as a loop variable in zsh; it is tied to `PATH`

13. Rules that protect production

1. **Know which database you are pointed at** before `db push`, seeds or e2e tests: print the host.
2. **Never put real secrets in a tracked file**, a commit message or a screenshot.
3. **Do not run the e2e tests, seeds or `db push` against production**. They are for development databases.
4. **Do not change `/api/mobile/v1` contracts** (routes, parameters, response shapes, the `{ error: { code, message } }` body). Add routes instead; the shipped mobile app depends on them.
5. **One network per deployment**: mainnet and testnet are never mixed in one environment.
6. **Deploy the worker when no pin batch is running** (`GET /health` → `jobs.running`). A job in hand is finished before the old container stops, but a very long batch can outlast the 120 s wait.
7. **Do not kill processes you didn't start.** A busy port is often someone's dev server.